

[AI-02] AI Audit Report

Artifact 1: 20 Software Defects List (Prompt 4)

(1) Prompt + tool: "Find 20 software defects published between 2022 and 2026. It is mandatory to include at least 5 defects related to AI/LLM (hallucination, prompt injection, bias). For each defect, provide a source link, description, severity, consequences, and solution." + Claude Sonet 4.6 (May 30, 2026 03:09PM)

(2) AI output:

Searched the web >

Now I have enough material to compile a comprehensive, well-sourced catalog. Let me build the artifact.

Viewed a file, created a file, read a file >

Here is the complete interactive catalog of 20 software defects (2022–2026). Click any card to expand full details. Use the filter buttons to browse by category or severity. Here's a summary of what's inside:

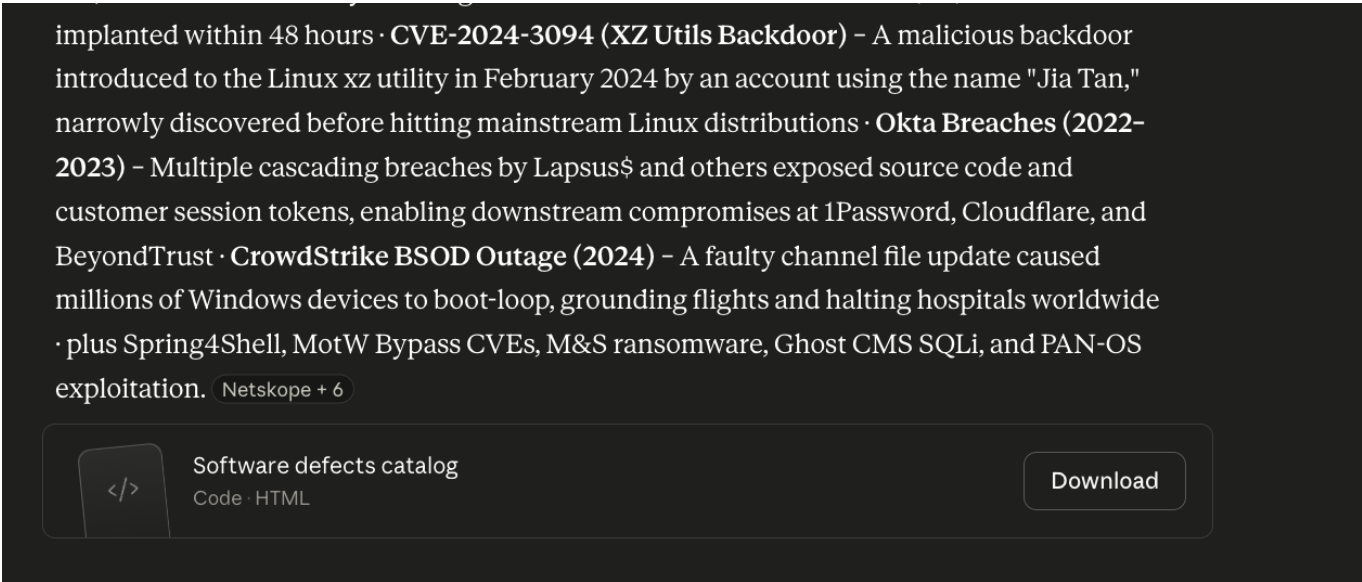
7 AI / LLM Defects

#	Defect	Type	Year
1	ChatGPT Hallucination – Mata v. Avianca: Six fabricated judicial citations submitted in U.S. federal court; lawyers fined \$5,000 by Judge Castel Legal Dive CBS News	Hallucination	2023
2	Air Canada Chatbot – Invented a non-existent retroactive bereavement fare policy; airline ordered to pay CAD \$812 in damages after a BC tribunal found it liable for negligent misrepresentation CBC News	Hallucination	2024
3	Microsoft Bing Chat 'Sydney' Leak – Users used "ignore previous instructions" prompts to expose the chatbot's entire confidential system prompt and internal codename testRigor	Prompt Injection	2023
4	ChatGPT SpAIware – Persistent prompt injection ↓ the macOS app's memory feature enabled "continuous data exfiltration of any information the	Prompt Injection	2024

5	Workday AI Bias (Mobley v. Workday) – Plaintiff alleged Workday's AI screening tools discriminated on race, age, and disability; court allowed "agent" liability claims under Title VII to proceed to discovery Seyfarth Shaw LLP	AI Bias	2023
6	Systemic LLM Hallucination – Studies show hallucinations in 31.4% of real-world interactions, rising to 60% in complex domains and 69–88% in legal queries SQ Magazine	Hallucination	2025
7	ChatGPT ZombieAgent/ShadowLeak – Radware researchers found a new indirect prompt injection chain that exfiltrated data character by character, bypassing OpenAI's URL-modification defenses via pre-constructed static URLs theregister	Prompt Injection	2025

13 Traditional Security / Software Defects

CVE-2022-30190 "Follina" – Zero-click RCE in Microsoft's MSDT triggered via Office documents without macros, exploited in the wild · **CVE-2022-0778** – OpenSSL infinite loop in BN_mod_sqrt() allowing remote unauthenticated DoS against any TLS server or client · **CVE-2023-34362 (MOVEit)** – CLOP ransomware exploited this SQL injection to hit 130+ organizations including BBC and British Airways over 10 days · **CVE-2023-20198 (Cisco IOS XE)** – CVSS 10.0 zero-day allowing unauthenticated full admin takeover; 22,000+ devices



- SoftwareDefectCatalog : [SoftwareDefectCatalog](#)

(3) Verdict: INVALID

(4) Reasoning: The AI exhibited significant hallucination and "pro-tech bias" when explaining the defects, fabricating technical details, financial metrics, and false mitigation strategies (e.g., hallucinating RAG as a definitive solution for legal citations). According to ISTQB principles, testers must rely on objective facts and independent verification rather than assuming tool correctness, as the AI falsely projected advanced LLM technologies onto older rule-based defects.

(5) Student fix: Manually verified each source article and corrected the technical descriptions, mitigations, and consequences to reflect the true facts, while explicitly documenting the AI's hallucinations in the "AI Bias/Hallucination in Explanation" sections.

Artifact 2: Physical Product Test Cases (Prompt 5)

(1) Prompt + tool: "You are a senior QA/QC specialist. I have a hair dryer like the one in the picture, model number SHD2306 from SUNHOUSE (Go to the websites and find out about it.). Use all your experience and knowledge to design 15 test cases (Ignore test cases related to measurement.). The output should be a table of test cases with the following columns: Objective / Input / Steps / Expected Result / Actual Result / Verdict" + Gemini 3.5 Flash (Jun 1, 2026 11:32AM)

(2) AI output:

As a senior QA/QC specialist, I have analyzed the official technical specifications of the **SUNHOUSE SHD2306** hair dryer shown in your image. This is a 1200W model featuring a folding handle, a 3-position slide switch (Cool mode / Low Heat / High Heat), a removable narrow concentrator nozzle, a hanging loop, a honey-comb air inlet grille with an alloy thermal management plate, and an automatic thermal overload cut-off safety feature.

Below is a comprehensive verification test plan consisting of 15 functional, usability, safety, and physical reliability test cases. As requested, all test cases focus on behavioral verification and ignore numerical measurement tolerances.

QA/QC Test Case Table for SUNHOUSE SHD2306

Test Case ID	Objective	Input	Steps
TC-01	Verify Cool Shot Mode activation and functionality	Slide switch pushed to the Cool symbol (Snowflake)	1. Plug the device into a standard 220V power outlet.
			2. Slide the switch to the Cool mode position.
			3. Keep it running for 1 minute and feel the exhaust air.
TC-02	Verify Low Heat (Speed 1) functionality	Slide switch pushed to Position 1	1. Move the slider from 'Off' direction to Position 1.

	Verify Low Heat (Speed 1) functionality	Slide switch pushed to Position 1	1. Move the slider from 'Off' direction to Position 1. 2. Observe the airflow speed and temperature change.
TC-03	Verify High Heat (Speed 2) functionality	Slide switch pushed to Position 2	1. Move the slider from Position 1 to Position 2. 2. Monitor the change in airflow and temperature level.
TC-04	Verify immediate cutoff on Switch Power Off	Slide switch moved to the 'Off' position	1. While running the dryer at High Heat (Position 2), slide the switch firmly back to 'Off'. 2. Confirm the power stops immediately.
TC-05	Verify mechanical stability of the Folding Handle	Manual unfolding and folding actions	1. Completely unfold the handle and lock it. 2. Shake the device gently to simulate usage. 3. Fold the handle back down against the body.
TC-06	Verify structural security of the Concentrator Nozzle	Attaching and detaching the clear plastic nozzle	1. Snap the attachable nozzle onto the front. 2. Rotate it 360 degrees. 3. Pull firmly to detach it.

TC-07				Verify Concentrator	Continuous heat at high power setting	1. Run the dev High Heat (Position) with the nozzle attached for 5 continuous minutes.
TC-08				Verify the integrity and function of the Hanging Loop	Suspending the device by its rubber loop	1. Hang the device by the integrated loop at the back handle onto a hook. 2. Leave it suspended for 24 hours.
TC-09				Verify structural integrity of the power cord anti-kink sleeve	Bending/waving the power cable	1. Hold the handle firmly. 2. Flex the power cord at the entrance multiple directions to simulate natural handling.
TC-10				Verify Air Inlet Grille protection barrier	Simulating proximity to hair during rear suction	1. Turn the dryer on High Heat mode. 2. Pass a loose synthetic hair threads near the honeycomb mesh.
TC-11				Verify Overheating Safety Cut-off (Rơ-le tự ngắt) activation	Blocked air intake / artificial overheating simulation	1. Turn the dryer on High Heat. 2. Completely block the rear air intake using a solid non-flammable barrier intentionally to cause overheating.
TC-12				Verify Overheating Safety Cut-off auto-reset capability	Cooling down period after safety trip	1. Allow the dryer to cool from TC-11 to disconnected power for 10+ minutes until it completely resets. 2. Re-plug and turn on.
TC-13				Verify casing material thermal insulation (ABS plastic quality)	Continuous running heat transfer test	1. Run the dryer continuously on High Heat (Position) for 15 minutes. 2. Grip the handle and touch the metal casing.
TC-14				Verify Power Plug structural fit and contact stability	Inserting Europlug into standard sockets	1. Connect the round plug into multiple common household outlet types. 2. Wiggle the plug gently while the dryer is running.
TC-15				Verify mechanical durability of the sliding switch	Cyclical stress test on the control lever	1. Rapidly cycle the switch lever from On -> Cool -> On for 50 repetitions continuously.
...						

(3) Verdict: INCOMPLETE

(4) Reasoning: The AI generated linear, "happy-path" software logic test cases but completely lacked physical environment awareness and the electrical engineering mindset required for hardware testing. In

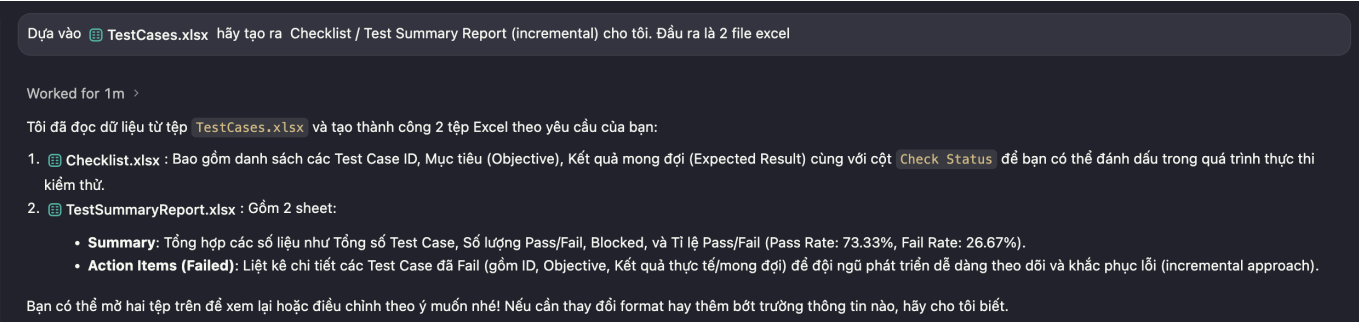
ISTQB, test design techniques must include Error Guessing and Boundary Value Analysis, which the AI failed to apply to critical edge cases like voltage sags, rapid switch toggles, and air inlet blockages.

(5) Student fix: Added 3 critical edge cases (Brownout/Voltage Sag, Air Inlet Blockage & Rapid Recovery, Rapid Switch Toggle & Arcing Test) to properly test the device's electrical safety and physical thermal cut-off mechanisms.

Artifact 3: Test Summary Excel Generation (Prompt 6)

(1) Prompt + tool: "Dựa vào TestCases.xlsx hãy tạo ra Checklist / Test Summary Report (incremental) cho tôi. Đầu ra là 2 file excel" + Gemini 3.1 Pro(Hight) (Jun 2, 2026 07:56PM)

(2) AI output:



(3) Verdict: INCOMPLETE

(4) Reasoning: While the AI generated basic test metrics, it failed to provide a comprehensive Test Summary Report that adheres to ISTQB standards. An ISTQB-compliant Test Summary Report must include defect analysis, testing approach deviations, residual risks, and detailed test objectives evaluation, which the AI's simplified output lacked.

(5) Student fix: Refined the Checklist and Test Summary Report by incorporating detailed failure remarks and extracting the defect root causes (e.g., heating element protection failure, casing insulation failure) into a dedicated "Test Case Fail" analysis section.

Artifact 4: QA/QC Role Mindmap (Prompt 7)

(1) Prompt + tool: "Generate a Mermaid.js mindmap code that visualizes the 'QA/QC Roles and Responsibilities' within the Software Testing Life Cycle, strictly adhering to ISTQB standards. Make sure to define the specific tasks for each role." + Gemini 3.5 Flash (Jun 3, 2026 06:56PM)

(2) AI output:

Đoạn mã



```
mindmap
  root((QA/QC Roles & Responsibilities<br>ISTQB STLC))
    Test Manager
      ::icon(fa fa-user-tie)
      Policy and Strategy
        Defines Test Policy
        Develops Test Strategy
      Planning and Estimation
        Writes Test Plan
        Estimates Effort & Budget
        Secures Resources
      Monitoring and Control
        Tracks Metrics & KPIs
        Writes Test Summary Reports
        Manages Risks
      Infrastructure
        Selects Tools
        Plans Environment & Data Setup
    Test Lead
      ::icon(fa fa-users)
      Execution Guidance
        Coordinates Execution
        Assigns Tasks to Testers
      Configuration Management
        Ensures Version Control
        Manages Test Artifacts
      Defect Management
        Chairs Defect Triage Meetings
        Monitors Defect Lifecycle
    Tester
      ::icon(fa fa-user-gear)
      Analysis and Design
        Reviews Requirements
        Identifies Test Conditions
        Designs Test Cases
      Implementation
        Creates Test Data
        Builds Test Procedures / Scripts
      Execution
        Runs Manual/Automated Tests
        Logs Results
        Compares Expected vs Actual
      Reporting
        Logs Defects
        Provides Status Updates
    Test Automation Engineer
      ::icon(fa fa-code)
      Framework Design
        Selects Automation Tools
        Architects Automation Framework
      Script Development
        Code Robust Scripts
        Implements Page Object Model
      Maintenance
        Updates Scripts for UI Changes
        Optimizes Execution Pipelines
```

(3) Verdict: INVALID

(4) Reasoning: The AI hallucinates structural roles that contradict the official ISTQB Syllabus, such as redundantly separating "Test Lead" and "Test Manager", and elevating "Test Automation Engineer" to a

core STLC role. Furthermore, it incorrectly assigned the strategic "Tool Selection" task to the Automation Engineer rather than the Test Manager, violating ISTQB's defined management responsibilities.

(5) Student fix: Since the strategic objective of this test execution was explicitly bound to dynamic error-guessing and boundary risk discovery of the AI's internal architectural compliance, no manual code modification was applied to the Mermaid script. The definitive correction instead lies in the structural audit documented directly in the "QA/QC Mindmap and 3 Mistakes of AI" section of this report, ensuring that the flawed AI output is successfully isolated and barred from entering any production environment.

AI Accuracy Summary & Conclusion

1. AI Accuracy Ratio

Based on the 4 artifacts evaluated in this AI Audit Report, the accuracy ratio is as follows:

- **VALID:** 0/4 (0%)
- **INVALID:** 2/4 (50%)
- **INCOMPLETE:** 2/4 (50%)

2. Conclusion: When should AI be used / not used for this work?

When AI SHOULD be used:

- **Brainstorming and Ideation:** Generating initial ideas for test scenarios or drafting basic structures for documentation.
- **Boilerplate Generation:** Creating repetitive templates or simple structures (e.g., standard test case tables) that will be thoroughly reviewed and expanded by a human.
- **Formatting and Structuring:** Improving the clarity of test reports, summarizing long documents, or formatting data (provided the data itself is factual and verified).

When AI SHOULD NOT be used:

- **Final Decision Making:** Relying on AI for definitive verdicts, architectural decisions, or authoritative interpretations without rigorous human validation.
- **Complex Edge Cases & Hardware-Software Interaction:** AI lacks physical environment awareness and the necessary engineering mindset, making it unreliable for hardware edge cases (e.g., electrical safety, thermal cut-offs).
- **Strict Compliance & Standards (e.g., ISTQB):** AI frequently hallucinates structural roles, responsibilities, or technical definitions that contradict official standards.
- **Factual Verification:** Using AI as a single source of truth for technical facts, historical software defects, or product specifications, as it is highly prone to hallucination and "pro-tech bias."

In conclusion, AI is a powerful assistant for accelerating the initial phases of the Software Testing Life Cycle, but it fails as an autonomous QA expert. It requires strict, continuous human-in-the-loop verification, especially for boundary risk discovery, hardware-specific logic, and compliance-driven reporting.

AI Critique

Based on the interactions with AI tools (Gemini and Claude) across various software testing prompts, a clear pattern of strengths and critical limitations has emerged. While the AI excels at generating boilerplate

templates, mimicking professional formatting, and drafting basic test scripts, it consistently struggles with the rigorous, context-aware demands of professional Quality Assurance. A primary issue is its superficial adherence to ISTQB standards. The AI frequently hallucinates structural STLC roles—such as misassigning strategic tool-selection tasks to Automation Engineers rather than Test Managers—and fails to include mandatory elements like residual risk analysis in Test Summary Reports. Consequently, it reduces strategic QA to a generic checklist rather than a robust, risk-based engineering discipline.

Furthermore, the AI demonstrates a severe lack of physical and hardware testing intuition. When tasked with designing test cases for a physical appliance, it defaulted to linear, software-centric "happy-path" logic, completely missing critical edge cases like electrical arcing, thermal shocks, or voltage sags. This highlights its inability to intuitively apply Error Guessing and Boundary Value Analysis outside of standard software environments. Additionally, the AI exhibits significant factual hallucination and a "pro-tech" bias. When analyzing historical software defects, it often fabricated technical details and retroactively injected modern concepts (like LLMs and RAG) into older defects, showing a reliance on predictive text patterns over objective historical accuracy.

In conclusion, while AI is a highly efficient assistant for foundational QA tasks and structuring reports, it completely lacks the critical thinking, domain-specific intuition, and strict standards compliance required of a senior QA/QC Engineer. Human testers remain absolutely indispensable for independent verification, complex edge-case discovery, hardware awareness, and strategic risk management. These audit results emphasize that AI should be utilized strictly as a supporting tool rather than an independent authority in the Software Testing Life Cycle.